Customer Support Automation with Amazon Lex Chatbot

Overview:

- To develop a customer support chatbot using Amazon Lex to enhance customer engagement, provide 24/7 support, and streamline the customer service process.

Description:

- This project involves the creation of an intelligent chatbot using Amazon Lex, a service for building conversational interfaces into any application using voice and text. The chatbot is designed to handle common customer inquiries, provide automated responses, and escalate complex issues to human agents when necessary.

Prerequisites:

We must have some basic knowledge on:

1. Chatbot - A chatbot is a software application designed to simulate human conversation through text or voice interactions over the internet.

2. Natural Language Processing (NLP) - The chatbot uses NLP to analyze the words and phrases in the message to understand the user's intent. NLP involves a series of techniques designed to allow computers to process, analyze, and generate human language in a meaningful way.

3. Intent - An intent represents an action that the user wants to perform. You create a bot to support one or more related intents. For example, you might create an intent to order pizzas and drinks. For each intent, you provide the following required information:

- Intent name - A descriptive name for the intent. For example, OrderPizza.

4. Sample utterances - How a user might convey the intent. For example, a user might say “Can I order a pizza” or “I want to order a pizza.”

How to fulfill the intent - How you want to fulfill the intent after the user provides the necessary information.

Data Flow:

1. User Interaction:

a. User Initiates a conversation with the chatbot using text or voice input through a web or mobile application.

b. Once the conversation is initiated then the application Sends the user's input to Amazon Lex.

2. Amazon Lex Processing:

c. When the user's input is sended to Amazon Lex, it receives the input and uses natural language understanding (NLU) to process the text or voice input.

d. Once the NLP is done, then Amazon Lex with the help ofIntent Recognition, AmazonLex identifies the user's intent based on the input.

e. Slot Filling: Lex collects any required information (slots) from the user to fulfill the intent.

3. Response Generation:

f. Fulfillment: Once Lex has gathered all necessary information, it either completes the action (e.g., book a ticket, order a product),response: Lex generates an appropriate response based on the fulfillment outcome.

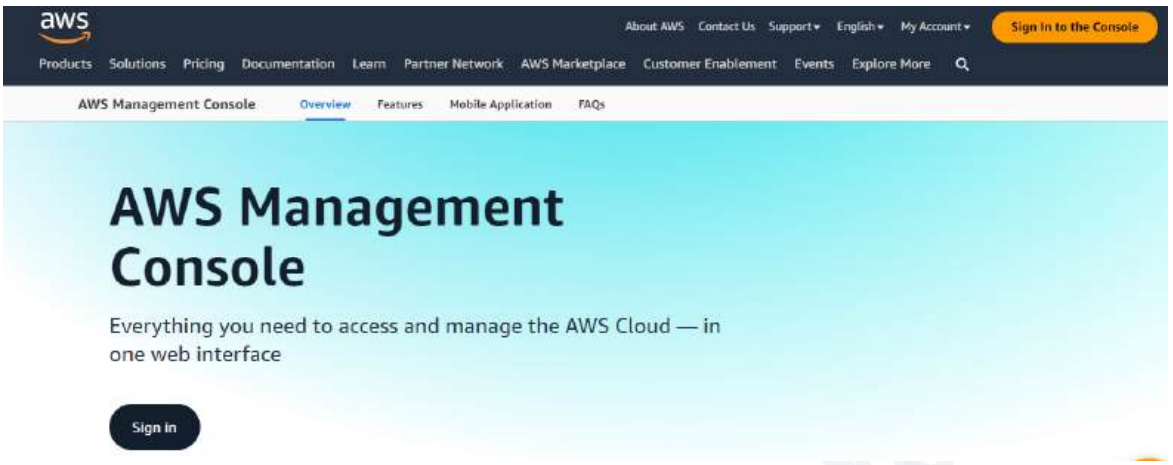
g. Application: Receives the response from Amazon Lex.User: Sees or hears the response from the chatbot.

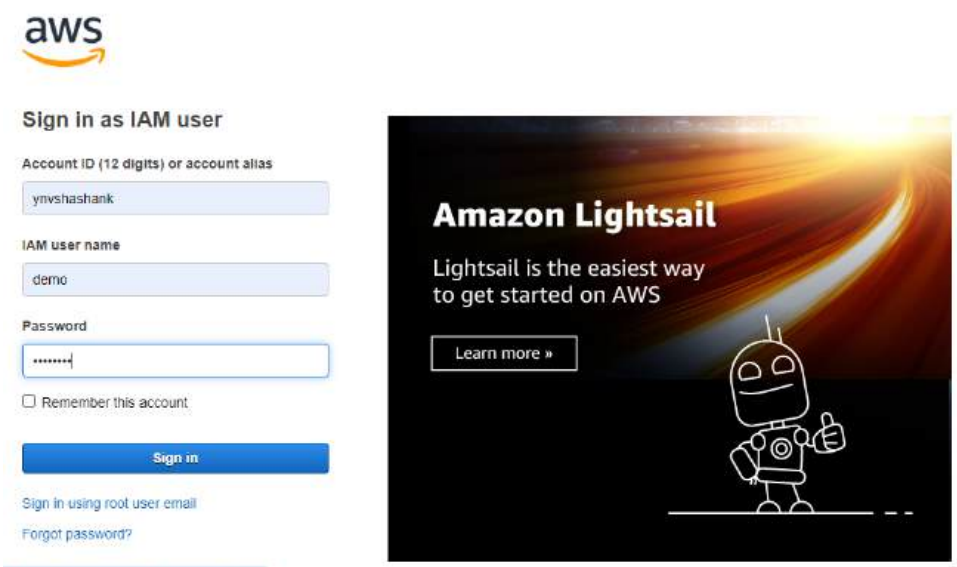
Rahul Lyadalla

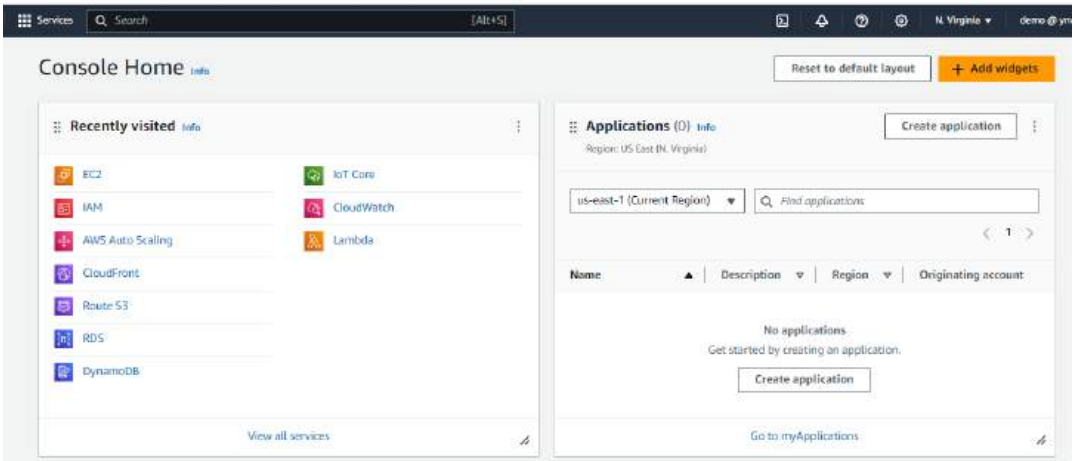
Steps:

Step 1: Login into AWS Console

1.1 Create an account and login into it by providing the credentials.

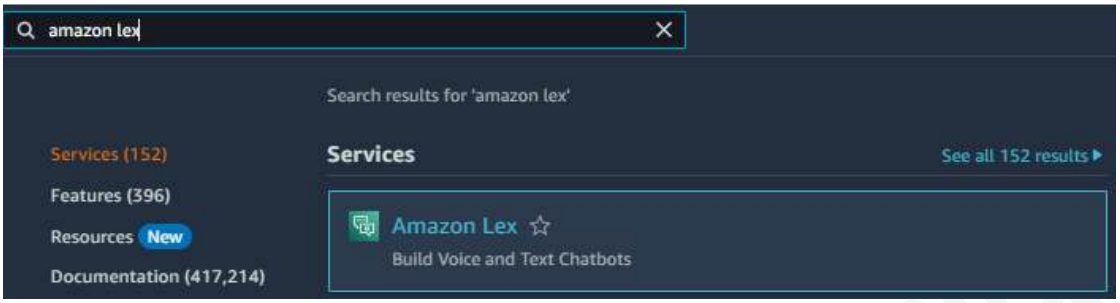




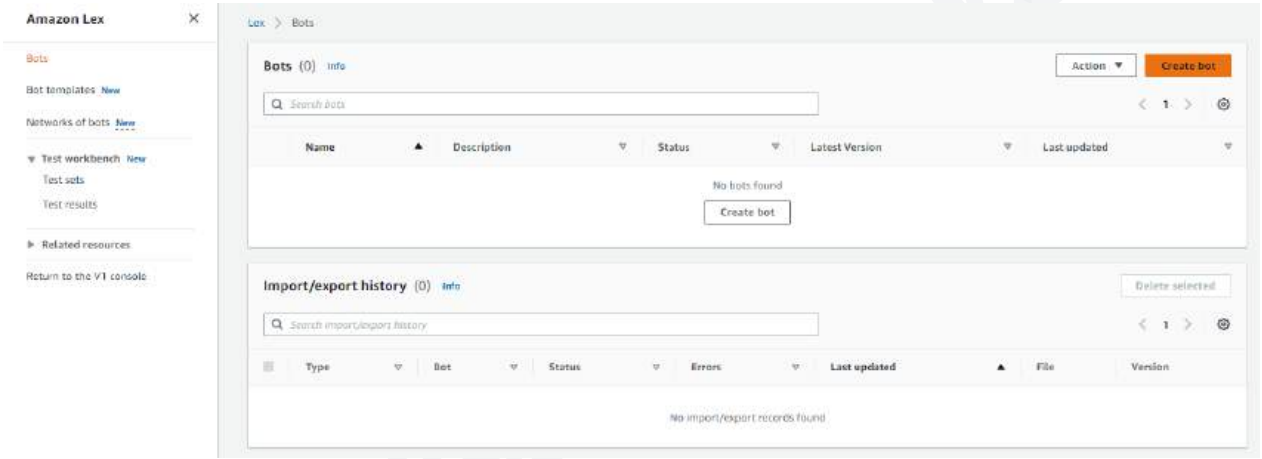


Step 2: Creating Chatbot using Amazon Lex

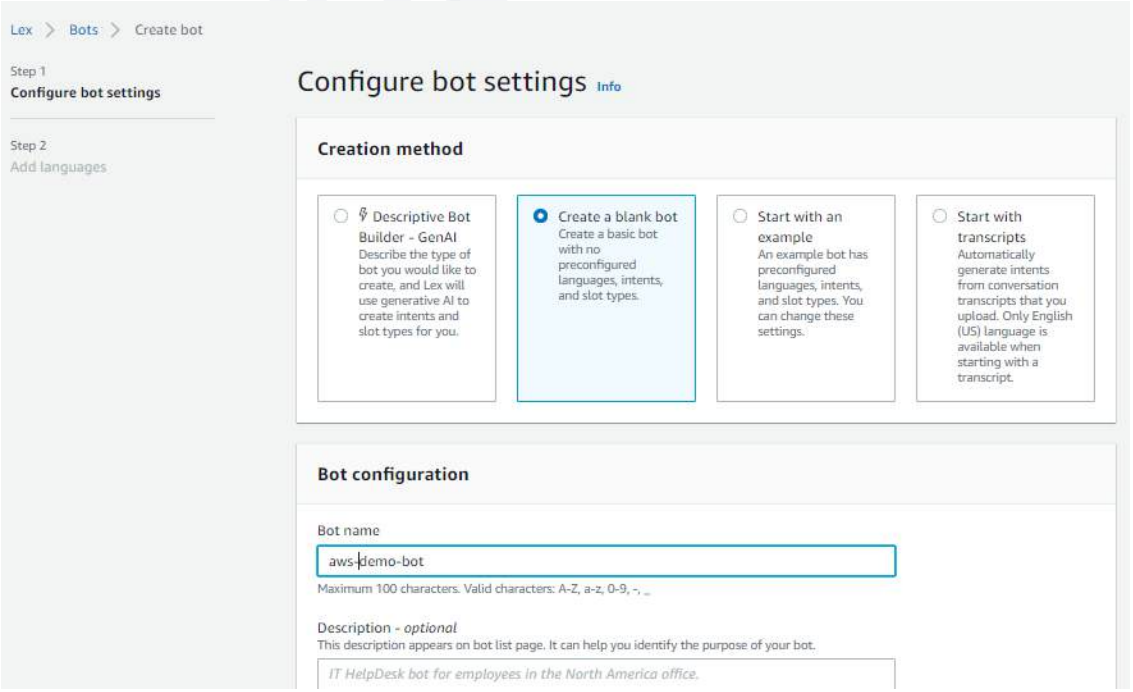
2.1 In Dashboard, search for **Amazon Lex,** then click on **Amazon Lex** to open it.



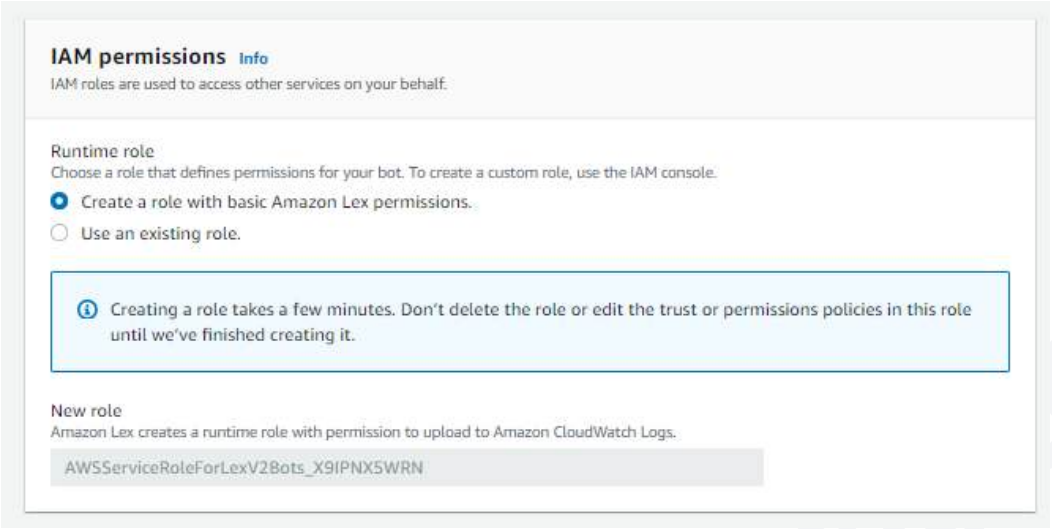
2.2 Now you will be forwarded to a bot space, then click on **create bot**.



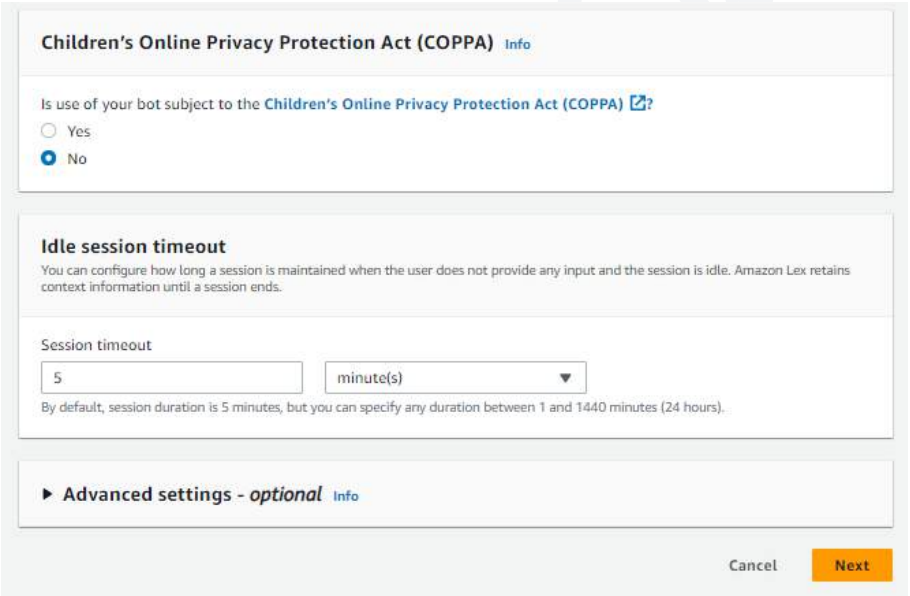
2.3 In **Configure bot settings,** you need to give the name of the bot.



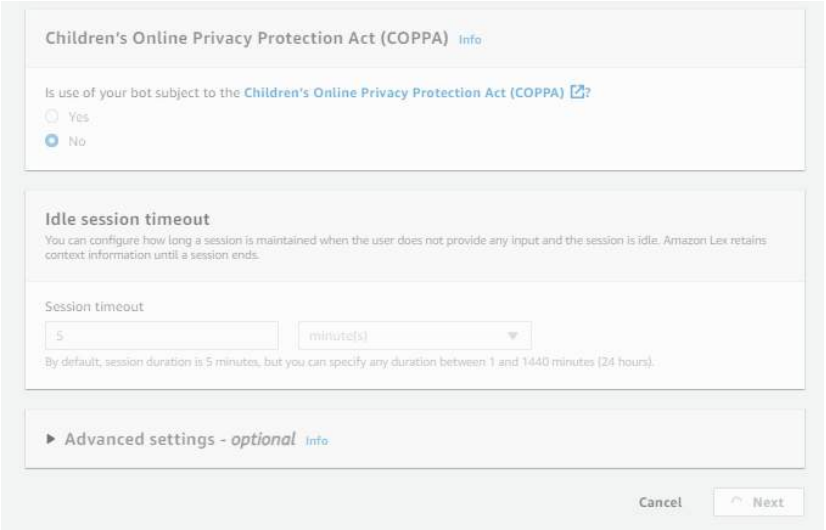
2.4 In IAM permissions Click on Create a Role with basic Amazon Lex permissions.



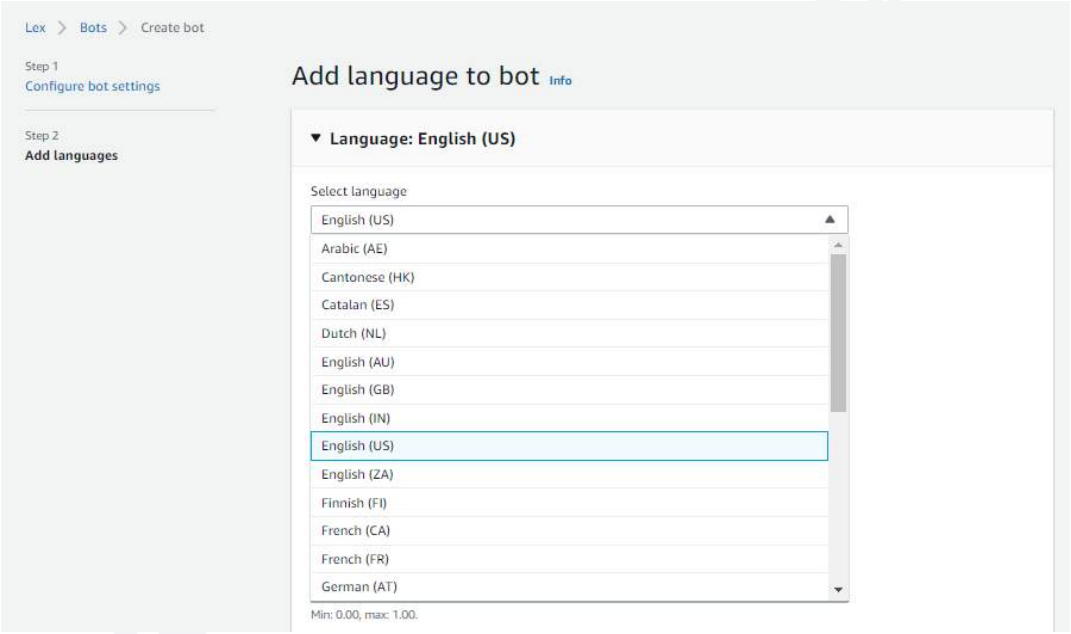
2.5 In **Children Online Privacy Protect Act (COPPA)**, Click on **NO**, as we are not going to store any personal data of the users, then Click on **Next**.

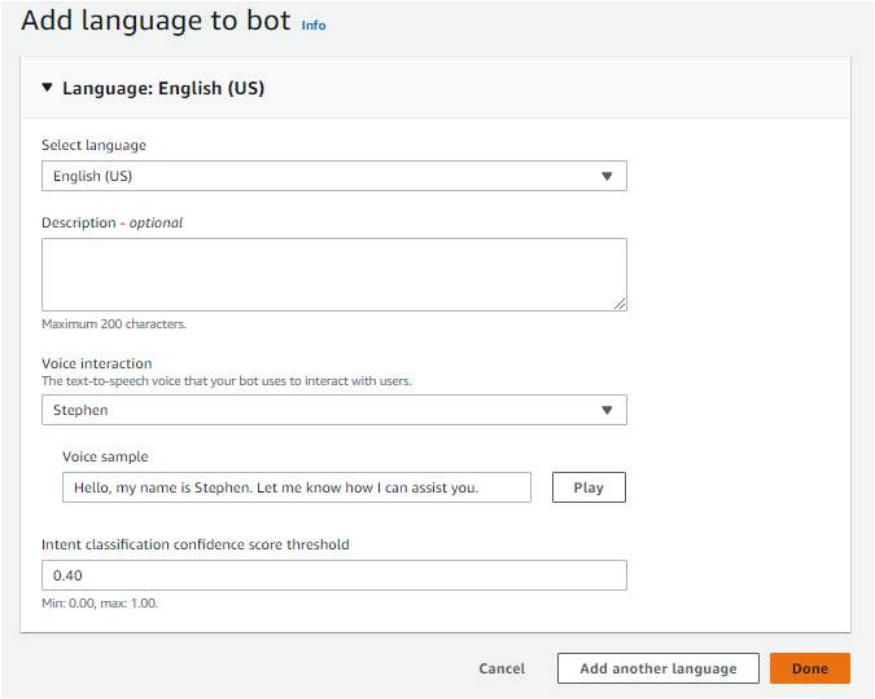


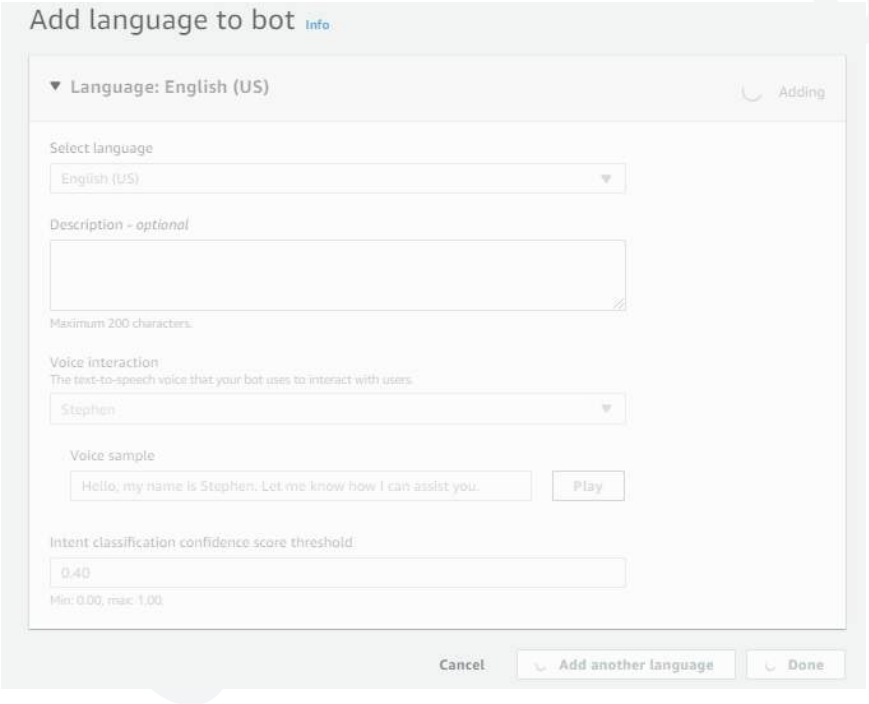
- It takes some time to load, please wait until it loads.



2.6 Now you have to select the language as English(US), and then you can select any name in voice interaction and you can check the voice sample by clicking on the play option and then Click on Done.




Add language to bot Info
Language: English (US)
Select language
English (US)
Description - optional
Maximum 200 characters.
Voice interaction
The text-to-speech voice that your bot uses to interact with users.
Stephen
Voice sample
Hello, my name is Stephen. Let me know how I can assist you.
Play
Intent classification confidence score threshold
0.40
Min: 0.00, max: 1.00.
Cancel
Add another language
Done


Add language to bot Info
Language: English (US)
Adding
Select language
English (US)
Description - optional
Maximum 200 characters.
Voice interaction
The text-to-speech voice that your bot uses to interact with users.
Stephen
Voice sample
Hello, my name is Stephen. Let me know how I can assist you.
Play
Intent classification confidence score threshold
0.40
Min: 0.00, max: 1.00.
Cancel
Add another language
Done

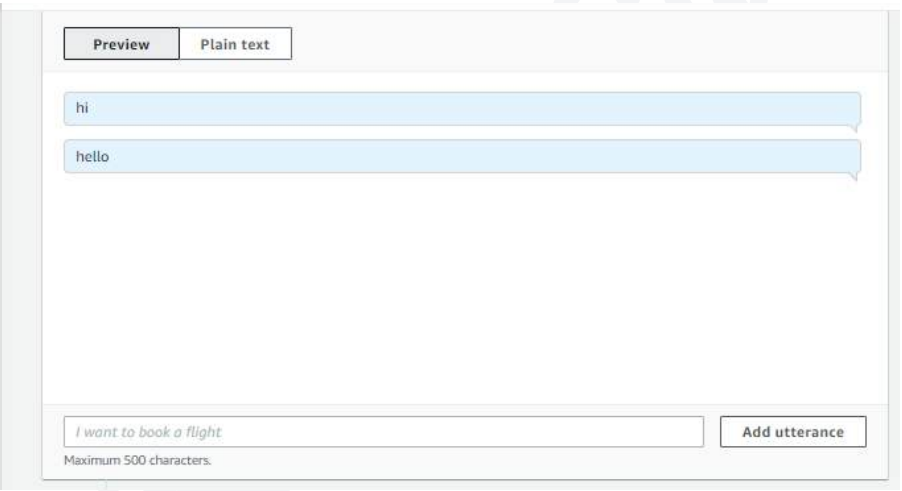
Successfully created bot: aws-demo-bot
Lex > Bots > Bot: aws-demo-bot > Versions > Version: Draft > All languages > Language: English (US) > Intents > Intent: NewIntent
Draft version
English (US)
Not built
English (US) has not built changes.
Build
Test

Step 3: Creation of Intents for chatbot

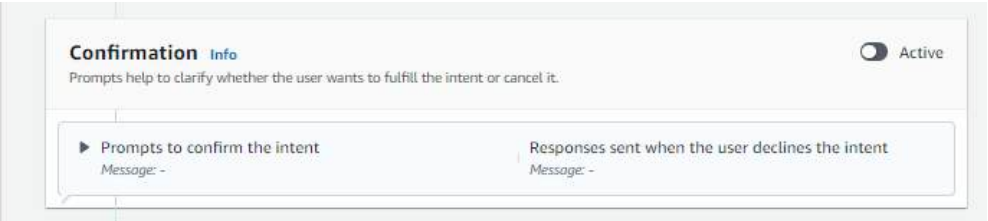
3.1 Now you will be forwarded to the Intent page, Intent is nothing but it is an action that the user wants to perform, for that you have to give an intent name .

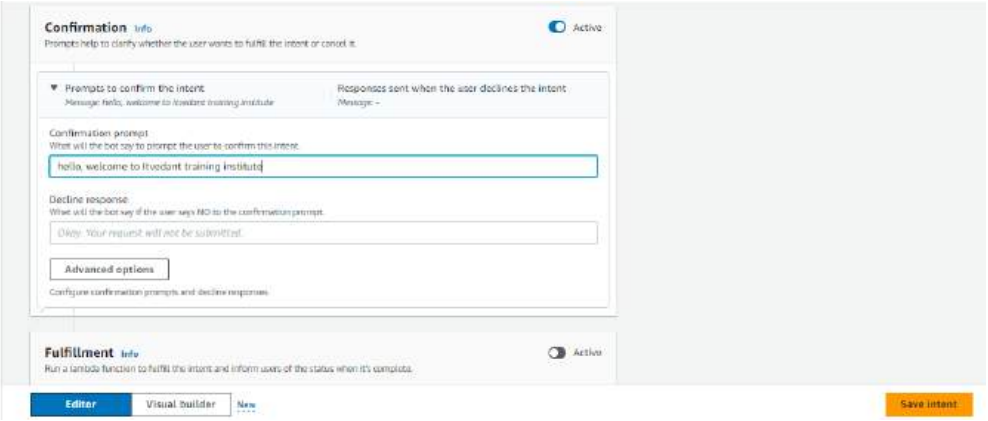
3.2 Now you have to go to the Sample Utterances and give some Utterances in the Add utterances Column, Utterances are nothing but the variation in the phrases that are expected outcomes.

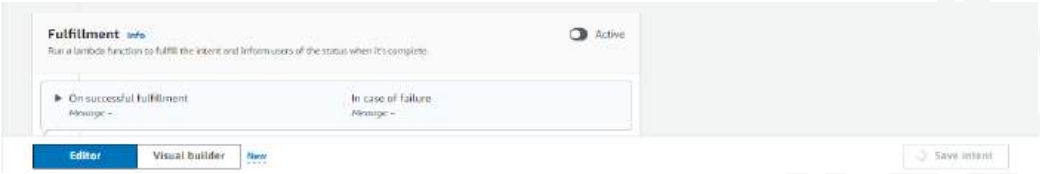
So to the question hello you have to add the variation of the phrases which are very close to that sentence, so add a few utterances by clicking on Add Utterances.



3.3 In **Conformation,** select the **Prompts to confirm the intent,** now you have to type the expected outcome or the expected answer for the given Intent Hello, then Click on the Save intent.





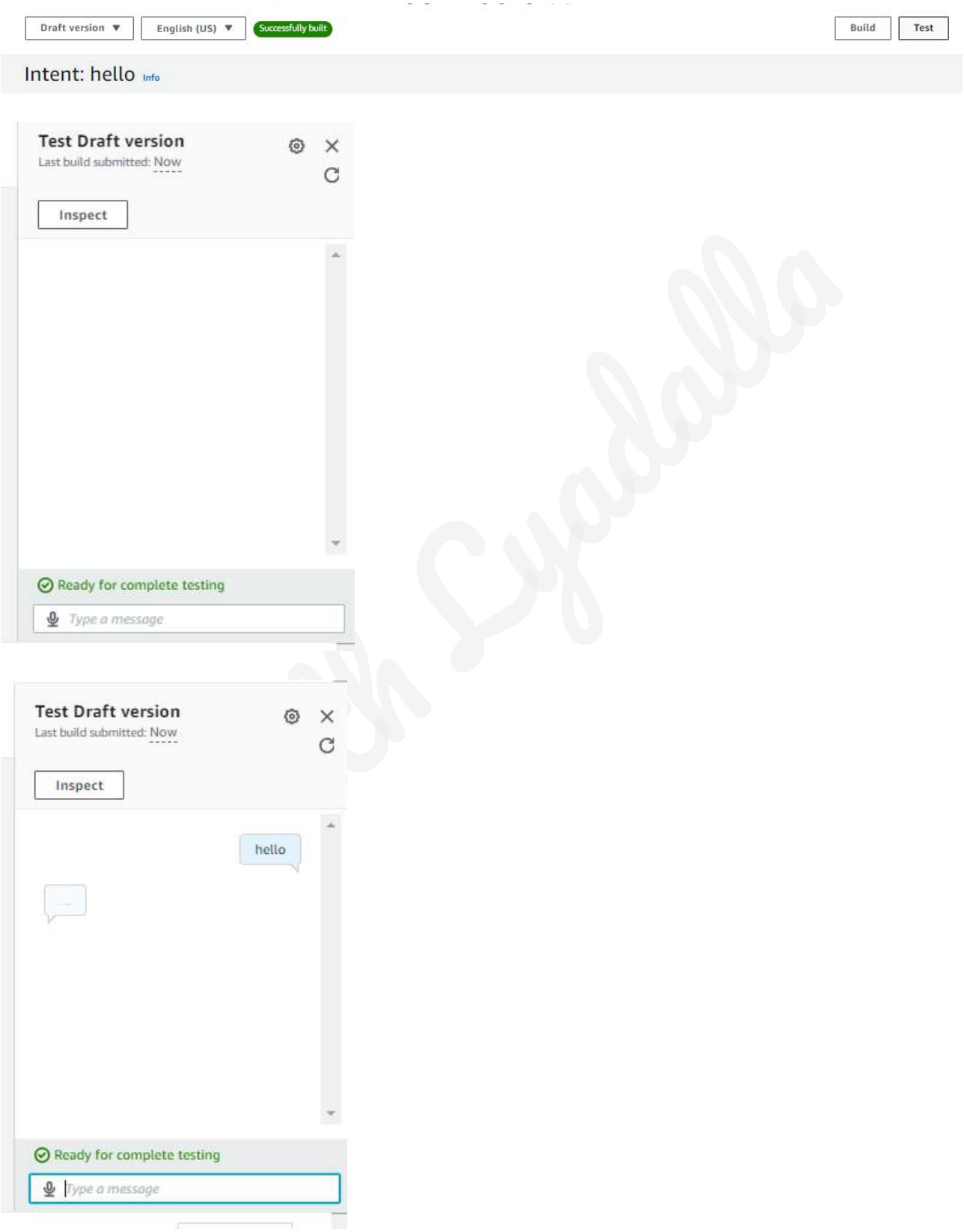


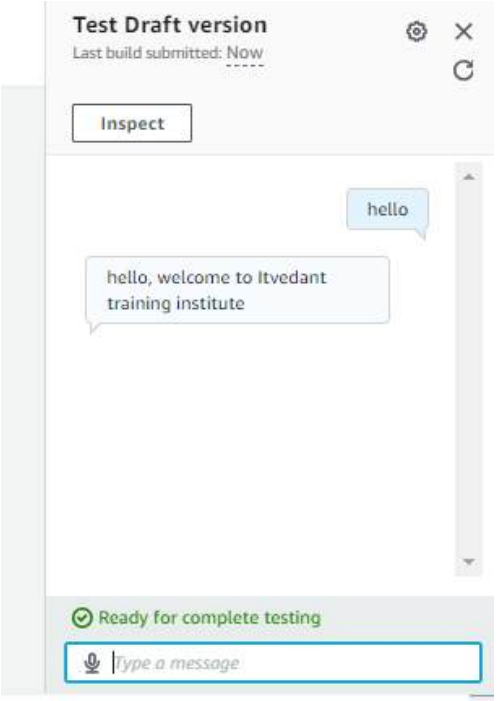
Step 4: Checking the Chatbot

4.1 If you want to check the chatbot, you can Click on **Build** and you have to wait for some time to build it.

4.2 After Building is completed, to Test the chatbot, now Click on Test, you will be directed to a chatbot, now you can test the chatbot by giving the

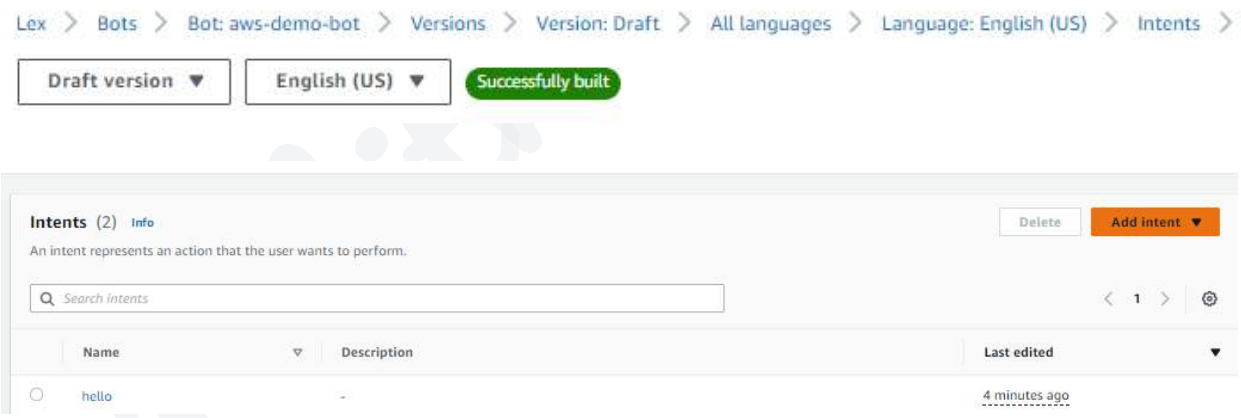
intent, what you have created, then you will be getting a desired output or response for your given question.



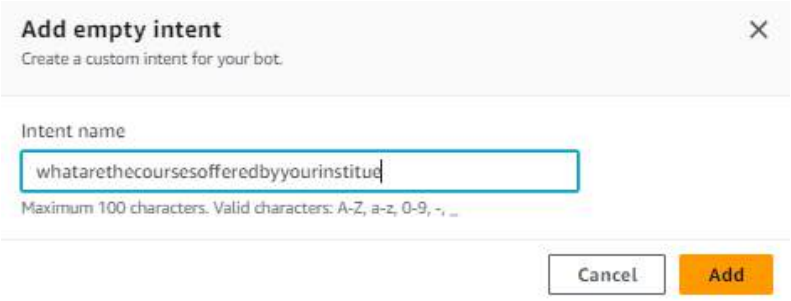


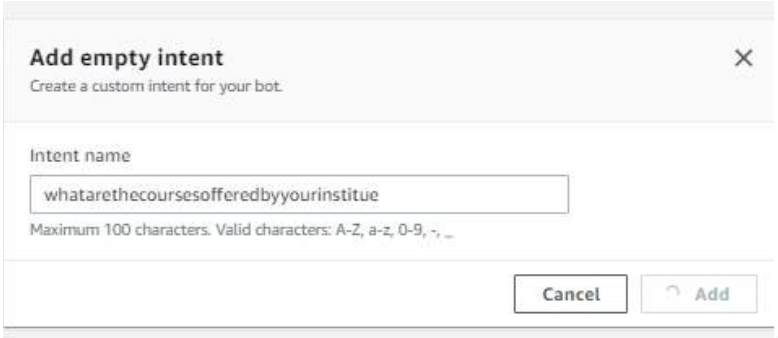
Step 5: Adding More Intents

5.1 To add more intents, Click on Intents, then Click on Add Intents, two options will be popping up, Click on Add empty intent.

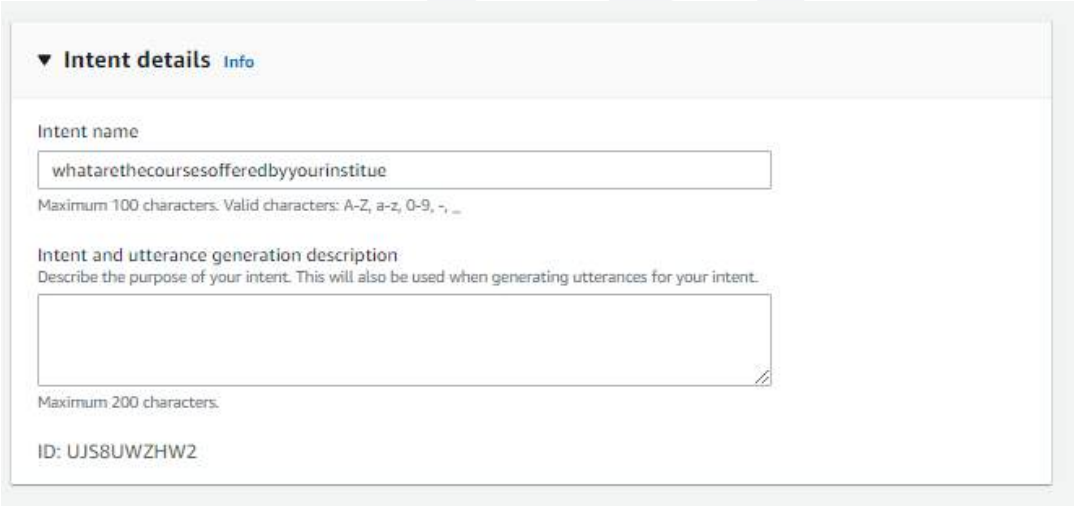


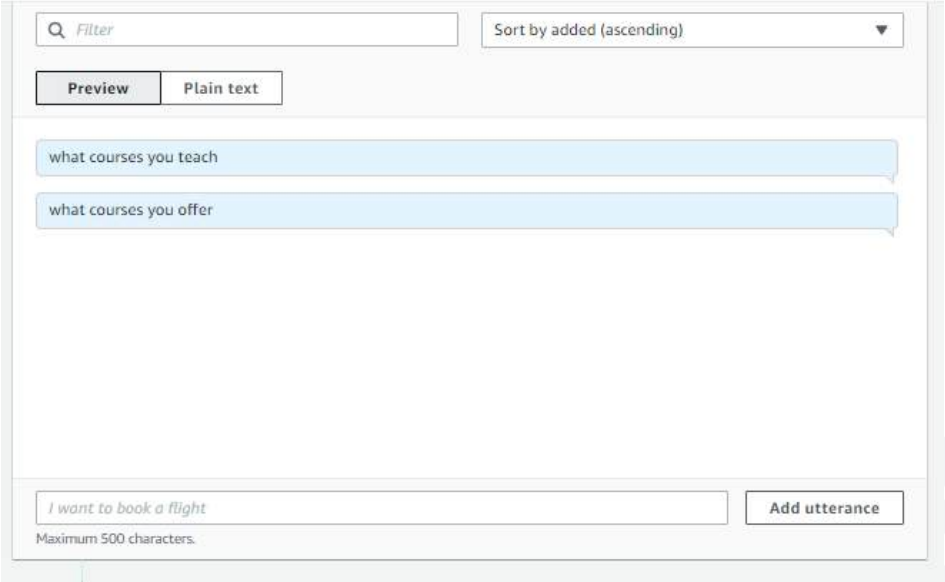
5.2 In Add empty intent, Give another intent name, then Click on Add.

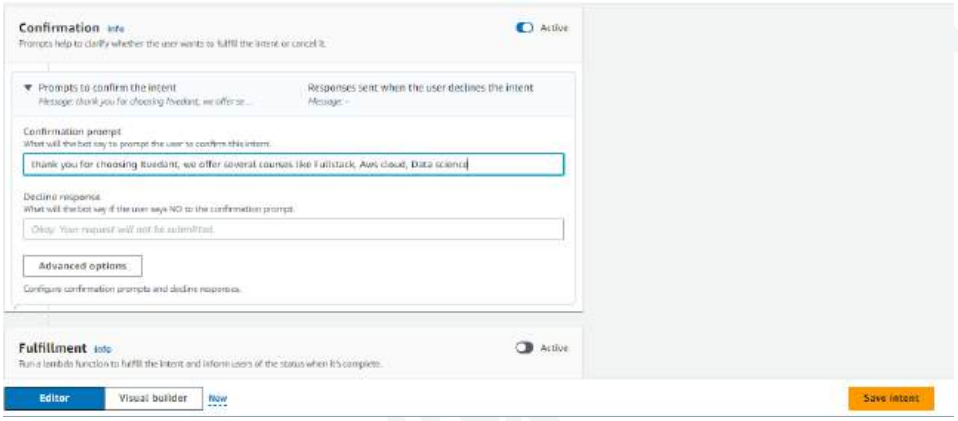


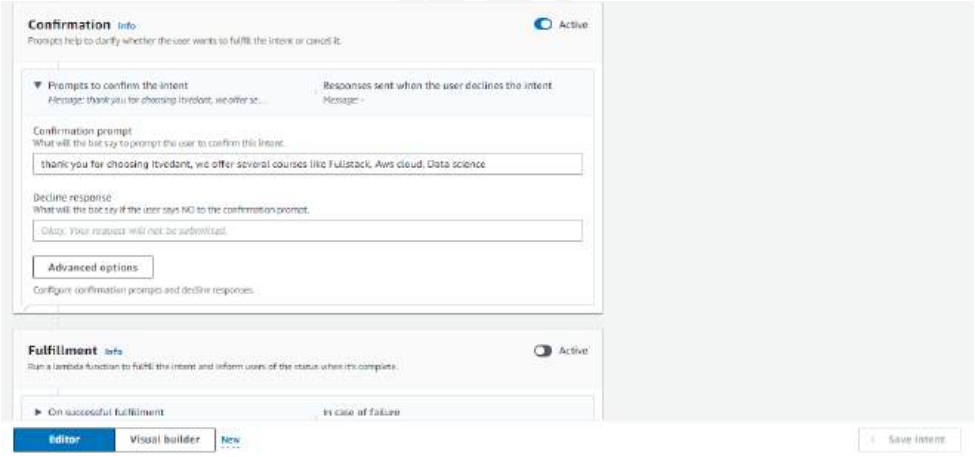


5.3 Now you will be directed to **intent details**,scroll down to **Add Utterances**, Now give a few Utterances which match your to intents and Click on **Save intent**.







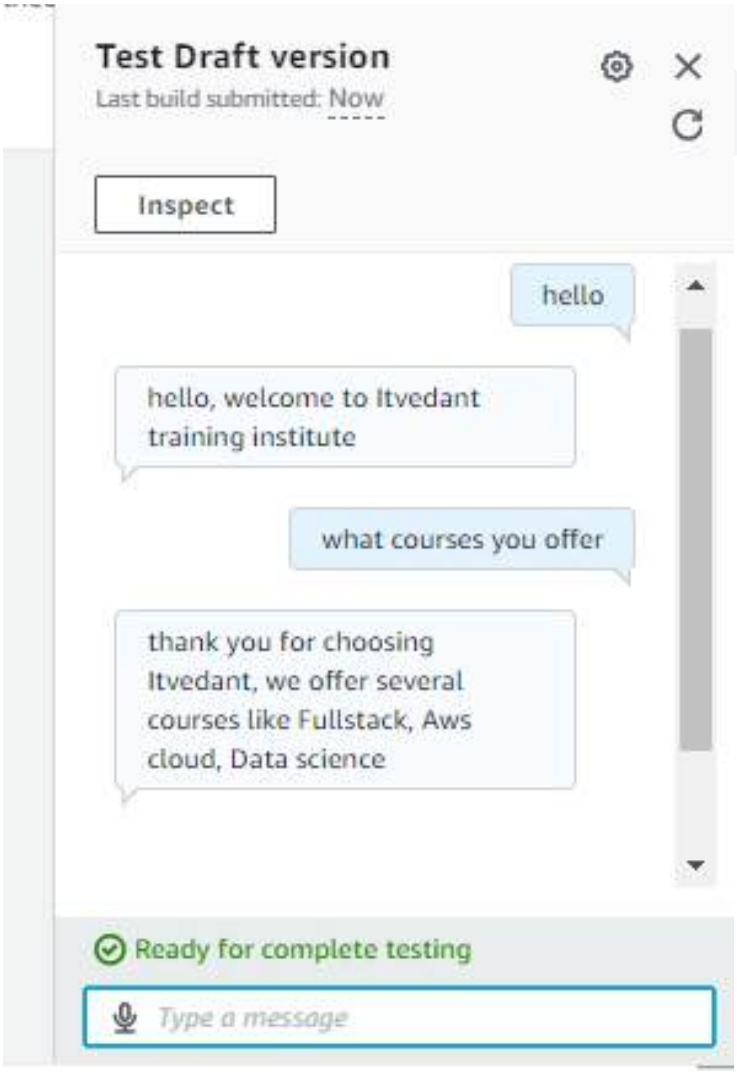


5.4 If you want to check the chatbot, you can Click on **Build** and you have to wait for some time to build it.

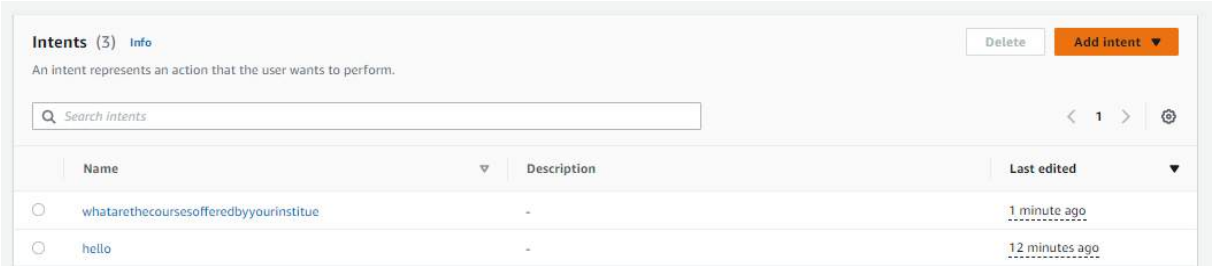
Building language English (US) in bot: aws-demo-bot. If your language contains external source slot types, the build might take longer to complete.
Lex > Bots > Bot: aws-demo-bot > Versions > Version: Draft > All languages > Language: English (US) > Intents > Intent: whatarethecoursesoff...
Draft version
English (US)
Building
Build
Test
Intent: whatarethecoursesofferedbyyourinstitue
Info

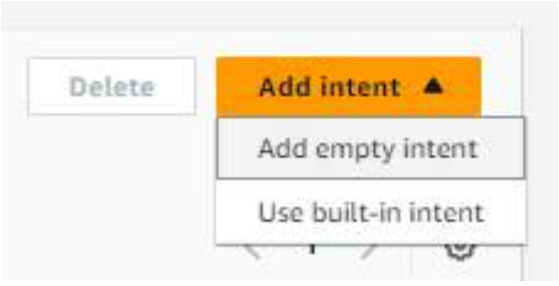
Successfully built language English (US) in bot: aws-demo-bot
Lex > Bots > Bot: aws-demo-bot > Versions > Version: Draft > All languages > Language: English (US) > Intents > Intent: whatarethecoursesoff...
Draft version
English (US)
Successfully built
Build
Test
Intent: whatarethecoursesofferedbyyourinstitue
Info

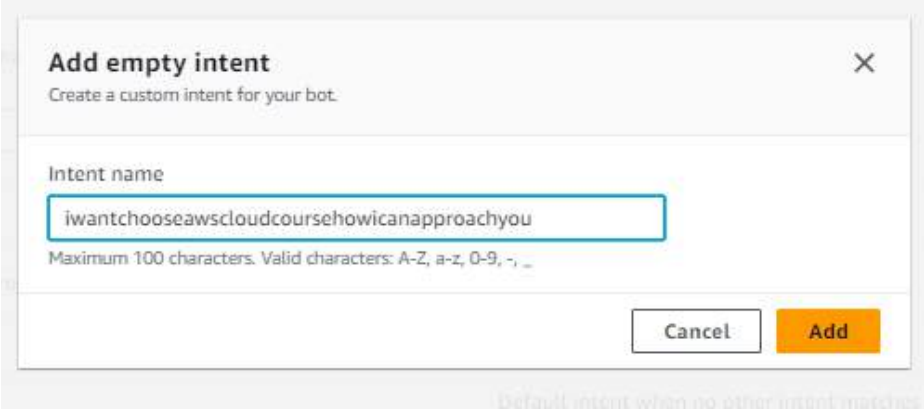
Build
Test

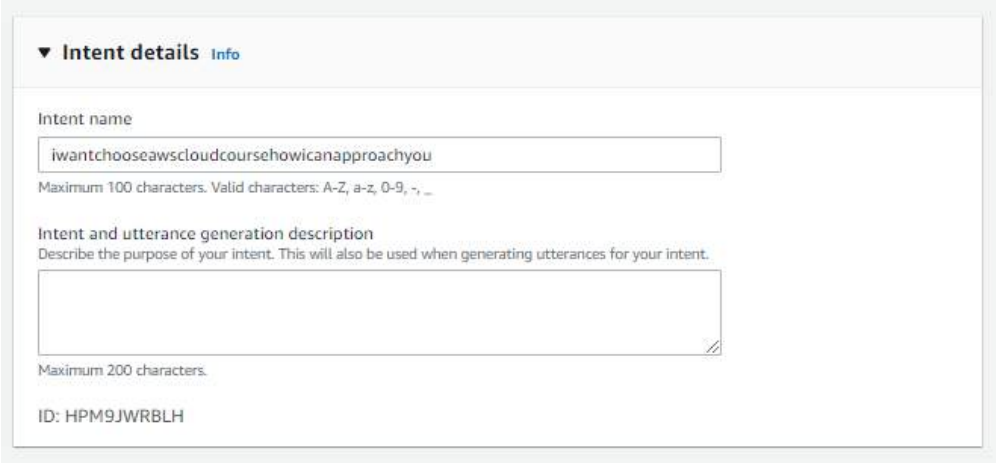

Test Draft version
Last build submitted: Now
Inspect
hello
hello, welcome to Itvedant training institute
what courses you offer
thank you for choosing Itvedant, we offer several courses like Fullstack, Aws cloud, Data science
Ready for complete testing
Type a message

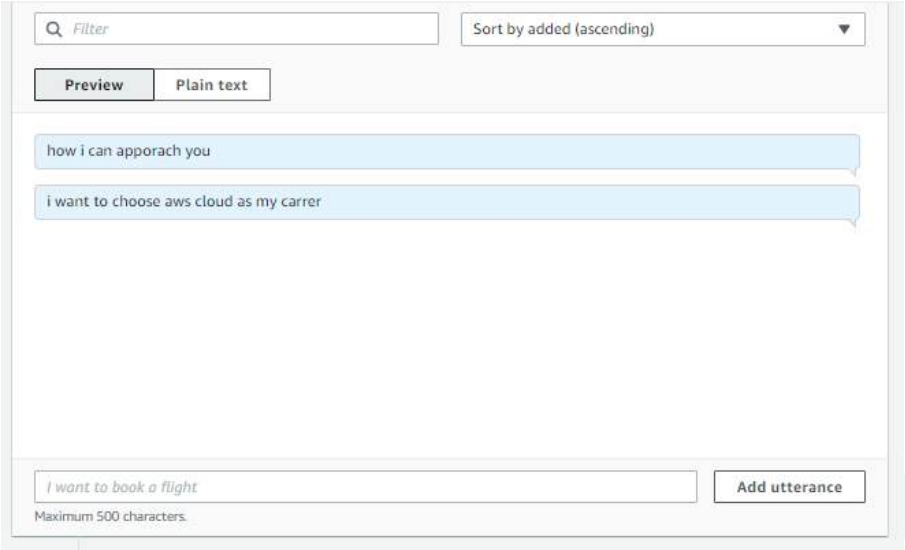
5.5 To Add more Intents, follow the above steps again.

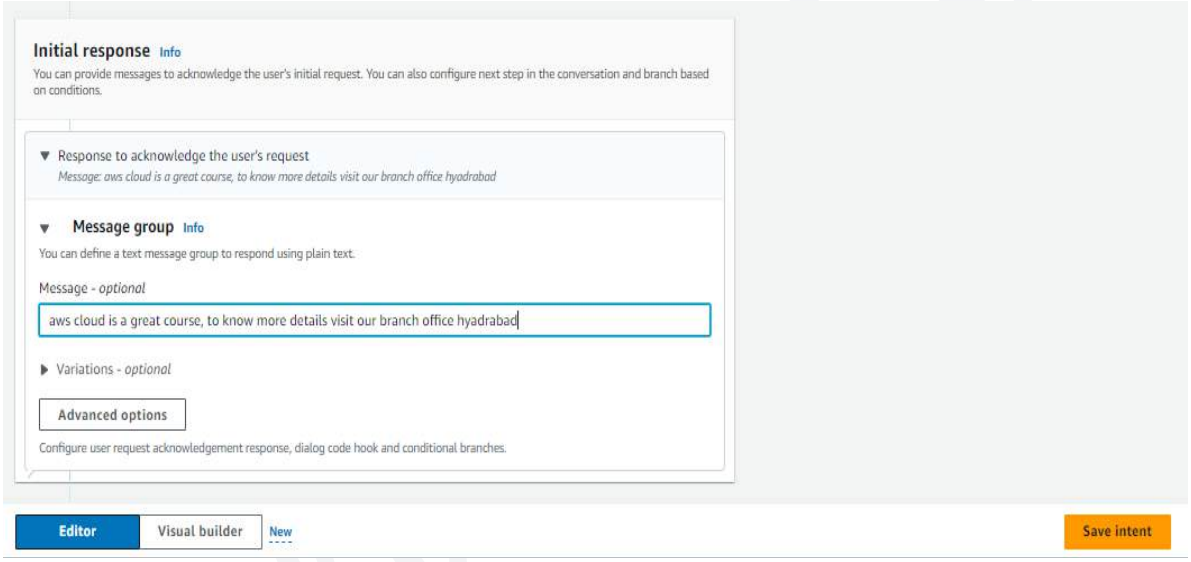






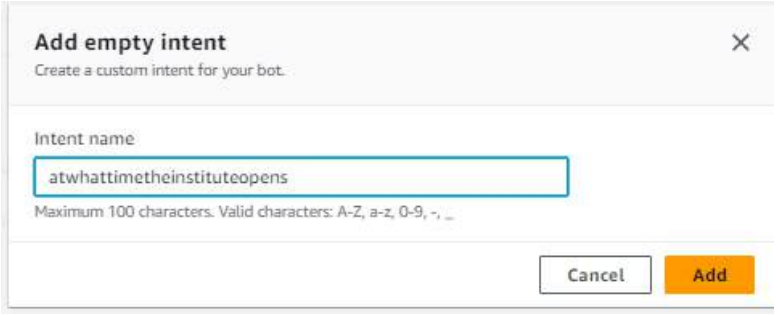


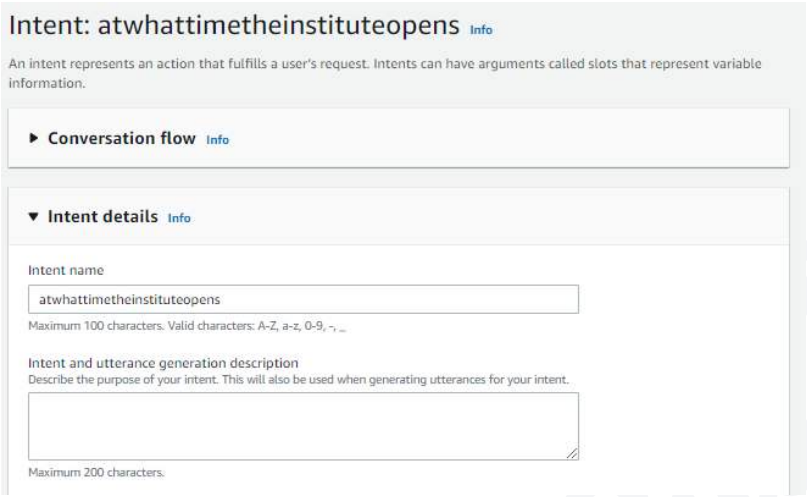

Filter
Sort by added (ascending)
Preview
Plain text
how i can apporach you
i want to choose aws cloud as my carrer
I want to book a flight
Add utterance
Maximum 500 characters.

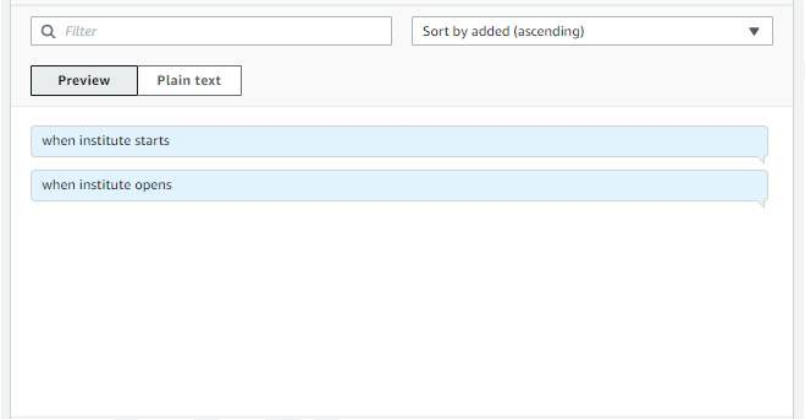

Initial response Info
You can provide messages to acknowledge the user's initial request. You can also configure next step in the conversation and branch based on conditions.
Response to acknowledge the user's request
Message: aws cloud is a great course, to know more details visit our branch office hyadrabad
Message group Info
You can define a text message group to respond using plain text.
Message - optional
aws cloud is a great course, to know more details visit our branch office hyadrabad
Variations - optional
Advanced options
Configure user request acknowledgement response, dialog code hook and conditional branches.
Editor
Visual builder
New
Save intent

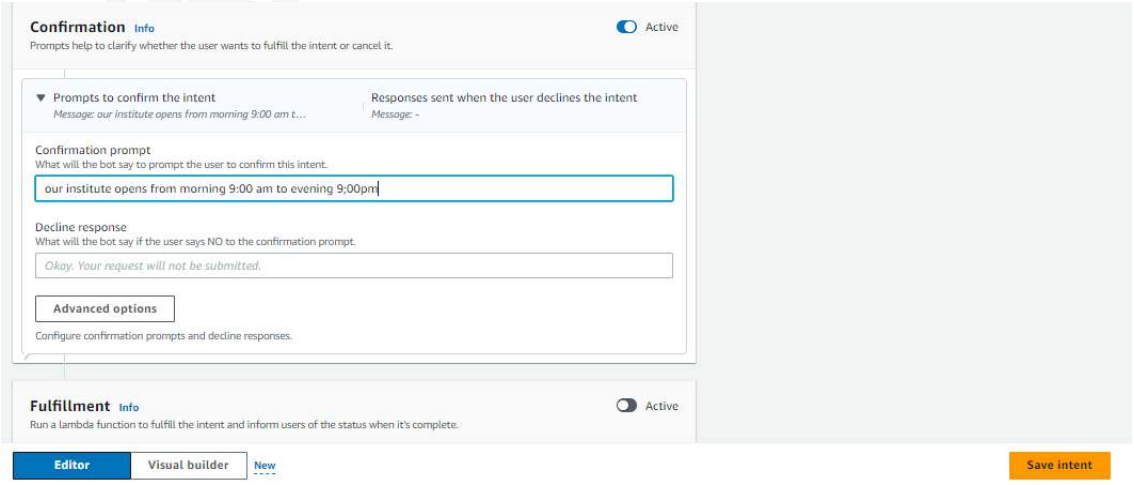
Building language English (US) in bot: aws-demo-bot. If your language contains external source slot types, the build might take longer to complete.
Lex > Bots > Bot: aws-demo-bot > Versions > Version: Draft > All languages > Language: English (US) > Intents > Intent: iwantchooseawscloudc...
Draft version
English (US)
Building
Build
Test
Intent: iwantchooseawscloudcoursehowicanapproachyou Info

Successfully built language English (US) in bot: aws-demo-bot
Lex > Bots > Bot: aws-demo-bot > Versions > Version: Draft > All languages > Language: English (US) > Intents > Intent: iwantchooseawscloudc...
Draft version
English (US)
Successfully built
Build
Test
Intent: iwantchooseawscloudcoursehowicanapproachyou Info


Add empty intent
Create a custom intent for your bot.
Intent name
atwhattimetheinstituteopens
Maximum 100 characters. Valid characters: A-Z, a-z, 0-9, -, _
Cancel
Add

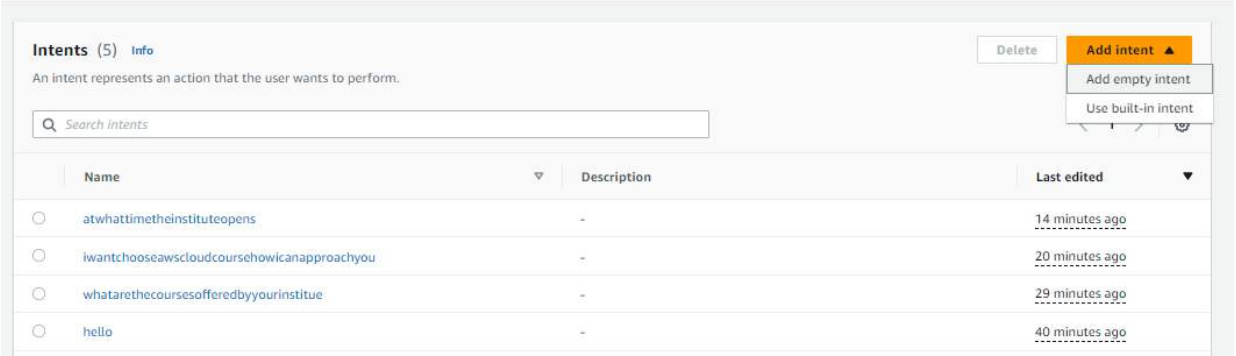

Intent: atwhattimetheinstituteopens Info
An intent represents an action that fulfills a user's request. Intents can have arguments called slots that represent variable information.
Conversation flow Info
Intent details Info
Intent name
atwhattimetheinstituteopens
Maximum 100 characters. Valid characters: A-Z, a-z, 0-9, -, _
Intent and utterance generation description
Describe the purpose of your intent. This will also be used when generating utterances for your intent.
Maximum 200 characters.

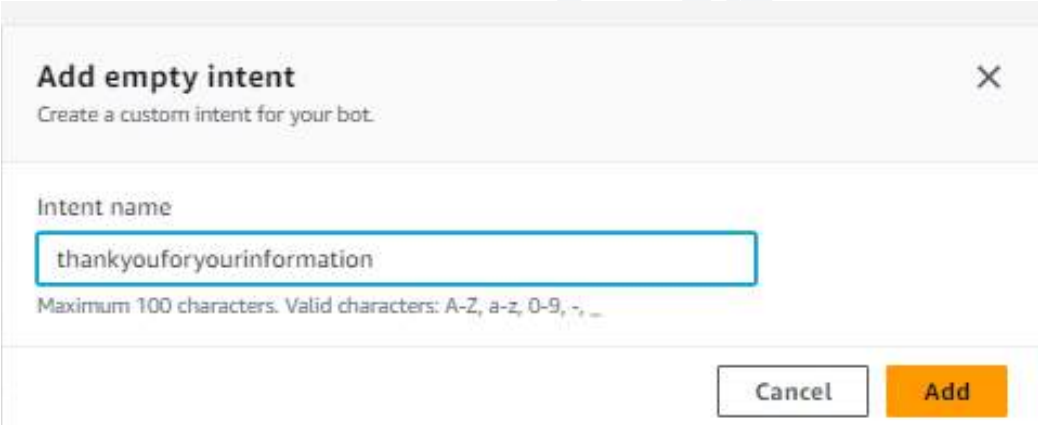

Filter
Sort by added (ascending)
Preview
Plain text
when institute starts
when institute opens


Confirmation Info
Active
Prompts help to clarify whether the user wants to fulfill the intent or cancel it.
Prompts to confirm the intent
Message: our institute opens from morning 9:00 am t...
Responses sent when the user declines the intent
Message: -
Confirmation prompt
What will the bot say to prompt the user to confirm this intent.
our institute opens from morning 9:00 am to evening 9:00pm
Decline response
What will the bot say if the user says NO to the confirmation prompt.
Okay. Your request will not be submitted.
Advanced options
Configure confirmation prompts and decline responses.
Fulfillment Info
Active
Run a lambda function to fulfill the intent and inform users of the status when it's complete.
Editor
Visual builder
New
Save intent

Building language English (US) in bot: aws-demo-bot. If your language contains external source slot types, the build might take longer to complete.
Lex > Bots > Bot: aws-demo-bot > Versions > Version: Draft > All languages > Language: English (US) > Intents > Intent: atwhattimetheinstitu...
Draft version
English (US)
Building
Build
Test
Intent: atwhattimetheinstituteopens
Info

Successfully built language English (US) in bot: aws-demo-bot
Lex > Bots > Bot: aws-demo-bot > Versions > Version: Draft > All languages > Language: English (US) > Intents > Intent: atwhattimetheinstitu...
Draft version
English (US)
Successfully built
Build
Test
Intent: atwhattimetheinstituteopens
Info


Intents (5) Info
An intent represents an action that the user wants to perform.
Delete
Add intent
Add empty intent
Use built-in intent
Search intents
Name
Description
Last edited
atwhattimetheinstituteopens
-
14 minutes ago
iwantchooseawscloudcoursehowicanapproachyou
-
20 minutes ago
whatarethecoursesofferedbyyourinstitue
-
29 minutes ago
hello
-
40 minutes ago


Add empty intent
Create a custom intent for your bot.
Intent name
thankyouforyourinformation
Maximum 100 characters. Valid characters: A-Z, a-z, 0-9, -, _
Cancel
Add

Intent: thankyouforyourinformation Info

An intent represents an action that fulfills a user's request. Intents can have arguments called slots that represent variable information.

▸ **Conversation flow** Info

▾ **Intent details** Info

Intent name

thankyouforyourinformation

Maximum 100 characters. Valid characters: A-Z, a-z, 0-9, -, _

Intent and utterance generation description

Describe the purpose of your intent. This will also be used when generating utterances for your intent.

Maximum 200 characters.

Filter | Sort by added (ascending)

Preview | Plain text

greetings,for your information

it is very useful information, thankyou

I want to book a flight | Add utterance

Maximum 500 characters.

Confirmation Info — Active

Prompts help to clarify whether the user wants to fulfill the intent or cancel it.

▾ Prompts to confirm the intent
Message: thank you, see you in institution and have a ...

Responses sent when the user declines the intent
Message: -

Confirmation prompt

What will the bot say to prompt the user to confirm this intent.

thank you, see you in institution and have a nice day

Decline response

What will the bot say if the user says NO to the confirmation prompt.

Okay. Your request will not be submitted.

Advanced options

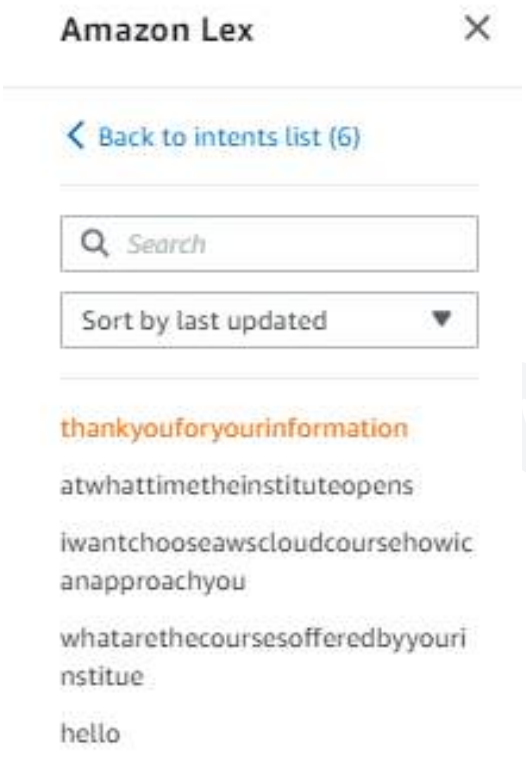
Configure confirmation prompts and decline responses.

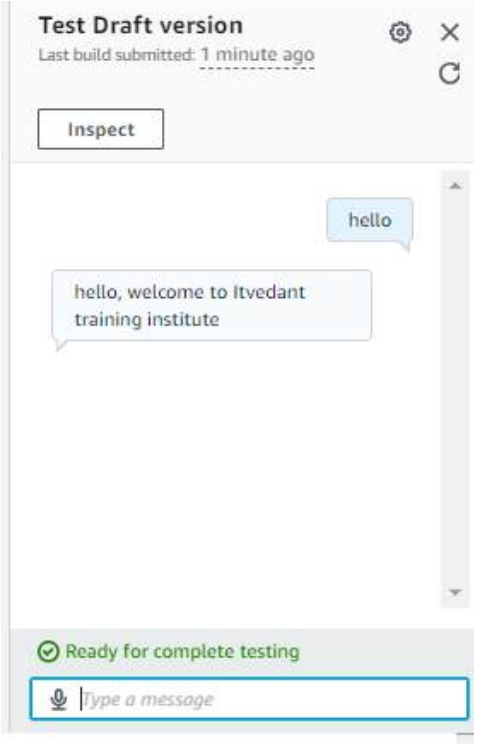
Fulfillment Info — Active

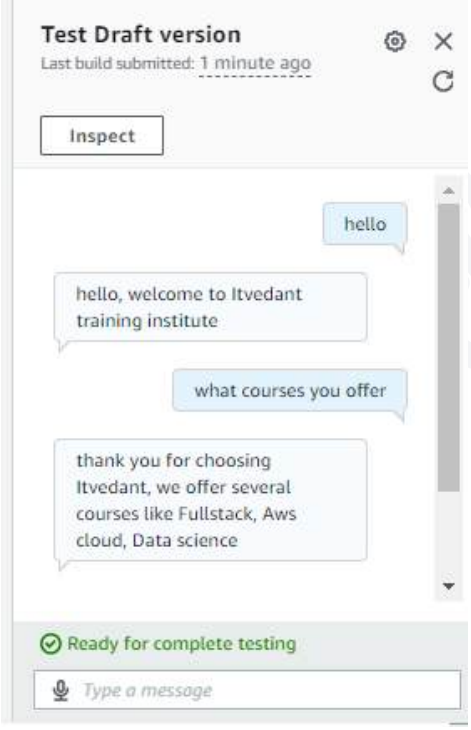
Run a lambda function to fulfill the intent and inform users of the status when it's complete.

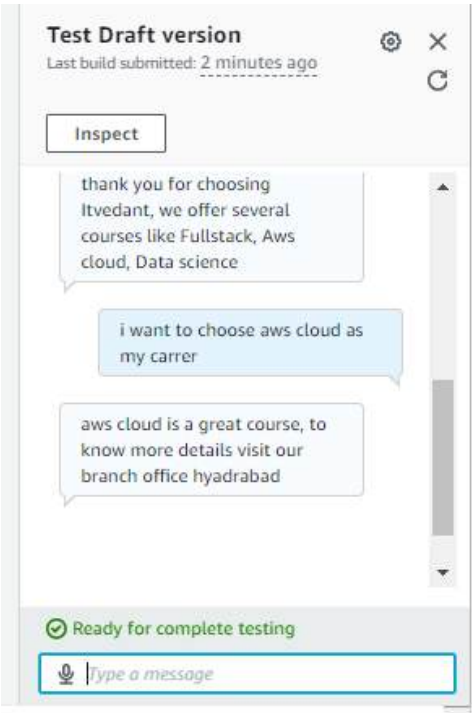
Editor | Visual builder New | Save intent

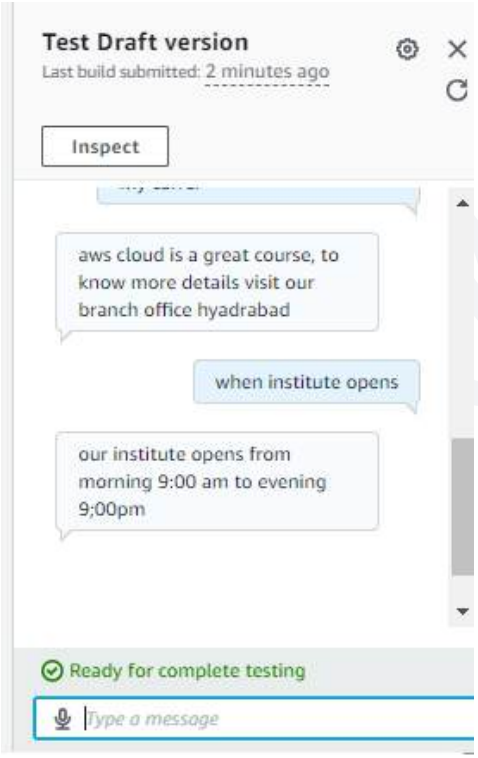
Step 6: Testing the Chatbot

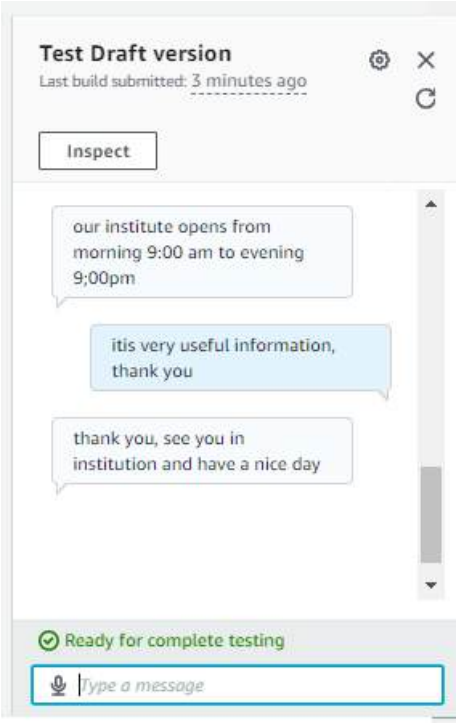



Test Draft version
Last build submitted: 1 minute ago
Inspect
hello
hello, welcome to Itvedant training institute
Ready for complete testing
Type a message


Test Draft version
Last build submitted: 1 minute ago
Inspect
hello
hello, welcome to Itvedant training institute
what courses you offer
thank you for choosing Itvedant, we offer several courses like Fullstack, Aws cloud, Data science
Ready for complete testing
Type a message


Test Draft version
Last build submitted: 2 minutes ago
Inspect
thank you for choosing Itvedant, we offer several courses like Fullstack, Aws cloud, Data science
i want to choose aws cloud as my carrer
aws cloud is a great course, to know more details visit our branch office hyadrabad
Ready for complete testing
Type a message


Test Draft version
Last build submitted: 2 minutes ago
Inspect
aws cloud is a great course, to know more details visit our branch office hyadrabad
when institute opens
our institute opens from morning 9:00 am to evening 9;00pm
Ready for complete testing
Type a message



Conclusion:

- With the help of Amazon Lex, which is a chatbot that can provide automated assistance, improve customer service, and enhance user experiences in various applications.

Rohith Gyadalla